Differential Equation

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Outline of Presentation

1. Introduction
2. Definition and Examples of Differential Equation
3. Classification of Differential Equations

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Intoduction

Differential equations are an important part of the calculus, the fundamentals of which are presented here. In developing the theory of differential equations in a systematic manner it is helpful to classify different types of equations.

There are two types of derivatives which usually are (and always can be) interpreted as rates. For example, the ordinary derivative $\frac{dy}{dx}$ is the rate of change of y with respect to x (independent variable), and the partial derivative $\frac{\partial y}{\partial x}$.

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Definition and Examples of Differential Equation

Definition

A differential equation is an equation which involves one or more derivatives, or differentials of an unknown function

Example 1

The following are examples of differential equations involving their respective unknown functions:

- $\frac{dy}{dx} = 3x^2$
- $\frac{\partial v}{\partial t} = h^2 \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right).$

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Classification of Differential Equations

Differential equations are classified as to:

- **Type**:We can place all differential equations into two types: ordinary differential equation (ODE) and partial differential equations (PDE).
- **Order**:The order of a differential equation is the order of the highest-ordered derivative appearing in the equation.
- **Degree**:The degree of the differential equation is the power (exponent) or the index that its highest ordered derivative is raised, if the equation is rationalized or cleared of fractions with regard to the dependent variable and its derivatives involve in it.

- **Linearity**: The conditions for a linear differential equation are as follows:
  1. The dependent variable and all its derivatives occur only in the first degree (or to the first power).
  2. No product of the dependent variable, say y , and/or any of its derivatives present
  3. No transcendental function (trigonometric, logarithmic or exponential) of the dependent variable and/or its derivatives occurs.

Example:

1. $3x^2y'' + 2\ln(x)y' + e^x y = 3x\cos x$.
   This is a second order linear ordinary differential equation
2. $4yy''' - x^3y' + \cos y = e^{2x}$.
   This is not a linear differential equation because of the $4yy'''$ and the $\cos y$ term

A **linear differential equation** is always in the **first degree** of the dependent variable.

Try

For each of the following, state whether the equations are ordinary or partial, linear or non-linear, and give its order and degree.

1. $\frac{d^2y}{dx^2} + kx^2 = 0$
2. $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = 0$
3. $y' + p(x)y = q(x)$
4. $\frac{\partial^2 w}{\partial t^2} = a^2 \frac{\partial^2 w}{\partial x^2}$

Solution to a DE

Definition

A solution of a differential equation is any function, say ϕ, satisfying the given differential equation on a specified interval, I .

Example 1:

The function $\phi = A\cos x + B\sin x$ is a solution of the differential equation $\dfrac{d^2\phi}{dx^2} + \phi = 0$.

Solution

$$\begin{aligned} \phi &= A\cos x + B\sin x \\ \frac{d\phi}{dx} &= -A\sin x + B\cos x \\ \frac{d^2\phi}{dx^2} &= -A\cos x - B\sin x \\ \Longrightarrow \frac{d^2\phi}{dx^2} + \phi &= (-A\cos x - B\sin x) + (A\cos x + B\sin x) = 0 \end{aligned}$$

Example 2:

Show that $y = \dfrac{1}{x^2 - 1}$ is a solution of $y' + 2xy^2 = 0$ 0n $I = (-1, 1)$ but not on any larger interval containing I.

Solution:

$$y = \frac{1}{x^2 - 1}, \quad \Longrightarrow \quad y' = \frac{-2x}{(x^2 - 1)^2}$$

$$y' + 2xy^2 = \frac{-2x}{(x^2 - 1)^2} + 2\left[\frac{1}{x^2 - 1}\right]^2 = 0.$$

Exercise

1. Show that $y = \ln x$ is a solution of $xy'' + y' = 0$ on $I = (0, \infty)$ but is not a solution on $I = (-\infty, \infty)$
2. Prove that $y = e^{-x} + \sin x$ is a solution of $\frac{d^2y}{dx^2} + y = 2e^{-x}$
3. Show that the two functions $y = (c^2 - x^2)^{\frac{1}{2}}$ and $-y = (c^2 - x^2)^{\frac{1}{2}}$ are both solutions of the equation $x + y\frac{dy}{dx} = 0$, $-c < x < c$.

Initial-value and Boundary-value Problems

- Initial-Value Problem is the differential equation along with subsidiary conditions on the unknown function and its derivatives, all given at the same value of the independent variable. The subsidiary conditions are initial conditions.
- Boundary-Value Problem is the differential equation along with subsidiary conditions on the unknown function and its derivatives, which are given at more than one value of the independent variable. The subsidiary conditions are boundary conditions.

Worked Example:

$y = c_1 e^{-x} + c_2 e^{3x}$ is a general solution of the differential equation: $y'' - 2y' + 3y = 0$. Determine the particular solution of the initial solution of the initial conditions $y = 3$ and $x = 0$ and $y' = 4$ when $x = 0$.

Solution:

$$y(0) = 3, \quad y'(0) = 4$$

$$\text{let } y = c_1 e^{-x} + c_2 e^{3x} \quad \text{when } x = 0,\ y = 3$$

$$\implies 3 = c_1 e^{(0)} + c_2 e^{3(0)} = c_1 + c_2$$

$$y' = -c_1 e^{-x} + 3c_2 e^{3x} \quad \text{when } x = 0,\ y' = 4,$$

$$\implies 4 = c_1 e^{(0)} + 3c_2 e^{3(0)}$$

$$\implies 4 = -c_1 + 3c_2, \text{ thus } c_1 = \frac{5}{4} \text{ and } c_2 = \frac{7}{4}$$

THANK YOU